1.1 Integer Exponents

For problems 1 - 4 evaluate the given expression and write the answer as a single number with no exponents.

1. $-6^2 + 4 \cdot 3^2$
2. $\dfrac{(-2)^4}{(3^2 + 2^2)^2}$
3. $\dfrac{4^0 \cdot 2^{-2}}{3^{-1} \cdot 4^{-2}}$
4. $2^{-1} + 4^{-1}$

For problems 5 - 9 simplify the given expression and write the answer with only positive exponents.

5. $\left(2w^4v^{-5}\right)^{-2}$
6. $\dfrac{2x^4y^{-1}}{x^{-6}y^3}$
7. $\dfrac{m^{-2}n^{-10}}{m^{-7}n^{-3}}$
8. $\dfrac{\left(2p^2\right)^{-3}q^4}{(6q)^{-1}p^{-7}}$
9. $\left(\dfrac{z^2y^{-1}x^{-3}}{x^{-8}z^6y^4}\right)^{-4}$

1.2 Rational Exponents

For problems 1 - 6 evaluate the given expression and write the answer as a single number with no exponents.

1. $36^{\frac{1}{2}}$
2. $(-125)^{\frac{1}{3}}$
3. $-16^{\frac{3}{2}}$
4. $27^{-\frac{5}{3}}$
5. $\left(\frac{9}{4}\right)^{\frac{1}{2}}$
6. $\left(\frac{8}{343}\right)^{-\frac{2}{3}}$

For problems 7 - 10 simplify the given expression and write the answer with only positive exponents.

7. $\left(a^3\, b^{-\frac{1}{4}}\right)^{\frac{2}{3}}$
8. $x^{\frac{1}{4}}\, x^{-\frac{1}{5}}$
9. $\left(\frac{q^3\, p^{-\frac{1}{2}}}{q^{-\frac{1}{3}}\, p}\right)^{\frac{3}{7}}$
10. $\left(\frac{m^{\frac{1}{2}}\, n^{-\frac{1}{3}}}{n^{\frac{2}{3}}\, m^{-\frac{7}{4}}}\right)^{-\frac{1}{6}}$

1.3 Radicals

For problems 1 - 4 write the expression in exponential form.

1. $\sqrt[7]{y}$
2. $\sqrt[3]{x^2}$
3. $\sqrt[6]{ab}$
4. $\sqrt{w^2v^3}$

For problems 5 - 7 evaluate the radical.

5. $\sqrt[4]{81}$
6. $\sqrt[3]{-512}$
7. $\sqrt[3]{1000}$

For problems 8 - 12 simplify each of the following. Assume that x, y and z are all positive.

8. $\sqrt[3]{x^8}$
9. $\sqrt{8y^3}$
10. $\sqrt[4]{x^7y^{20}z^{11}}$
11. $\sqrt[3]{54x^6y^7z^2}$
12. $\sqrt[4]{4x^3y}\ \sqrt[4]{8x^2y^3z^5}$

For problems 13 - 15 multiply each of the following. Assume that x is positive.

13. $\sqrt{x}\,(4-3\sqrt{x})$
14. $(2\sqrt{x}+1)\,(3-4\sqrt{x})$
15. $\left(\sqrt[3]{x}+2\ \sqrt[3]{x^2}\right)\left(4-\sqrt[3]{x^2}\right)$

For problems 16 - 19 rationalize the denominator. Assume that x and y are both positive.

16. $\dfrac{6}{\sqrt{x}}$

17. $\dfrac{9}{\sqrt[3]{2x}}$

18. $\dfrac{4}{\sqrt{x}+2\sqrt{y}}$

19. $\dfrac{10}{3-5\sqrt{x}}$

1.4 Polynomials

For problems 1 - 10 perform the indicated operation and identify the degree of the result.

1. Add $4x^3 - 2x^2 + 1$ to $7x^2 + 12x$
2. Subtract $4z^6 - 3z^2 + 2z$ from $-10z^6 + 7z^2 - 8$
3. Subtract $-3x^2 + 7x + 8$ from $x^4 + 7x^3 - 12x - 1$
4. $12y\left(3y^4 - 7y^2 + 1\right)$
5. $(3x + 1)\left(2 - 9x^2\right)$
6. $\left(w^2 + 2\right)\left(3w^2 + w\right)$
7. $\left(4x^6 - 3x\right)\left(4x^6 + 3x\right)$
8. $3\left(10 - 4y^3\right)^2$
9. $\left(x^2 + x - 2\right)\left(3x^2 - 8x - 7\right)$
10. Subtract $3\left(x^2 + 1\right)^2$ from $6x^3 - 9x^2 - 13x - 4$

1.5 Factoring Polynomials

For problems 1 - 4 factor out the greatest common factor from each polynomial.

1. $6x^7 + 3x^4 - 9x^3$
2. $a^3b^8 - 7a^{10}b^4 + 2a^5b^2$
3. $2x\left(x^2+1\right)^3 - 16\left(x^2+1\right)^5$
4. $x^2\left(2-6x\right) + 4x\left(4-12x\right)$

For problems 5 & 6 factor each of the following by grouping.

5. $7x + 7x^3 + x^4 + x^6$
6. $18x + 33 - 6x^4 - 11x^3$

For problems 7 - 15 factor each of the following.

7. $x^2 - 2x - 8$
8. $z^2 - 10z + 21$
9. $y^2 + 16y + 60$
10. $5x^2 + 14x - 3$
11. $6t^2 - 19t - 7$
12. $4z^2 + 19z + 12$
13. $x^2 + 14x + 49$
14. $4w^2 - 25$
15. $81x^2 - 36x + 4$

For problems 16 - 18 factor each of the following.

16. $x^6 + 3x^3 - 4$
17. $3z^5 - 17z^4 - 28z^3$
18. $2x^{14} - 512x^6$

1.6 Rational Expressions

For problems 1 - 3 reduce each of the following to lowest terms.

1. $\dfrac{x^2-6x-7}{x^2-10x+21}$
2. $\dfrac{x^2+6x+9}{x^2-9}$
3. $\dfrac{2x^2-x-28}{20-x-x^2}$

For problems 4 - 7 perform the indicated operation and reduce the answer to lowest terms.

4. $\dfrac{x^2+5x-24}{x^2+6x+8} \cdot \dfrac{x^2+4x+4}{x^2-3x}$
5. $\dfrac{x^2-49}{2x^2-3x-5} \div \dfrac{x^2-x-42}{x^2+7x+6}$
6. $\dfrac{x^2-2x-8}{2x^2-8x-24} \div \dfrac{x^2-9x+20}{x^2-11x+30}$
7. $\dfrac{\dfrac{3}{x+1}}{\dfrac{x+4}{x^2+11x+10}}$

For problems 8 - 12 perform the indicated operations.

8. $\dfrac{3}{x-4}+\dfrac{x}{2x+7}$
9. $\dfrac{2}{3x^2}-\dfrac{1}{9x^4}+\dfrac{2}{x+4}$
10. $\dfrac{x}{x^2+12x+36}-\dfrac{x-8}{x+6}$
11. $\dfrac{1}{x^2-13x+42}+\dfrac{x+1}{x-6}-\dfrac{x^2}{x-7}$
12. $\dfrac{x+10}{(3x+8)^3}+\dfrac{x}{(3x+8)^2}$